

Project Documentation

1. Introduction

Project Title: House Rent Portal

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2. Project Overview

Purpose

The HouseRent Portal is a web-based single-page application built on the MERN stack. The primary goal of this platform is to create a secure, transparent, and direct peer-to-peer ecosystem that connects property owners with verified tenants. By establishing direct communication pipelines, the portal completely eliminates the exploitation of middlemen and high brokerage fees, which typically delay and complicate the rental search process.

Features

Role-Based Sign-Up & Login: Secure authentication using JSON Web Tokens (JWT) and Bcrypt.js password hashing, routing users dynamically to their respective dashboards.

Admin Verification Pipeline: A dedicated moderation queue where administrators review and approve listings before they are made public, preventing fake and duplicate listing scams.

Direct Booking Requests: Renters can submit rental requests directly to the property owners, enabling quick, zero-brokerage negotiations.

Interactive Status Badges: Live status updates (Pending, Approved, Declined) displayed on dashboards so tenants can track their application status in real-time.

Dynamic Budget & Location Filters: Highly responsive search filters to help users easily discover properties matching their exact criteria.

3. Technology Stack

We chose a modern, high-performance stack to ensure fast client-side rendering and scalable database operations:

Frontend: React.js (Vite Engine) styled with utility-first Tailwind CSS.

Backend: Node.js with the Express.js framework for robust RESTful API routing.

Security: JSON Web Tokens (JWT) for stateless sessions and Bcrypt.js for securing sensitive user data.

Database: MongoDB Atlas (Cloud NoSQL Database) for flexible data modeling and fast query performance.

4. System Requirements

Hardware Requirements

CPU: Dual-Core Intel Core i3 / AMD Ryzen 3 minimum (Quad-Core recommended).

RAM: Minimum 8 GB RAM (to seamlessly run both React front-end and Node API servers simultaneously).

Disk Space: 20 GB of free space for project files, dependency packages (node_modules), and assets.

Software Requirements

Operating System: Windows 10/11, macOS, or Linux.

Runtime: Node.js LTS Engine (v16.x or higher) and NPM package manager (v8.x or above).

IDE: Visual Studio Code (VS Code) with extensions like ESLint and Prettier.

5. Data Flow Structures

User Authentication Flow

User Input → Express API → Bcrypt Hash → MongoDB Saved

Upon authentication, the backend generates and returns a signed secure JWT to the client to manage access.

Property Submission Flow

Owner Form Input → Client Validation → Route Controller → DB Document Creation (Status: 'Pending')

To maintain safety, new listings default to a "Pending" state and are hidden from search lists until an admin approves them.

Booking Pipeline

Renter Clicks 'Book Now' → Booking Created (Status: 'Pending')
→ Owner Dashboard Syncs → Owner Decision (Approve/Decline)
→ DB State Updated

Instant state hooks refresh dashboards automatically when states transition.